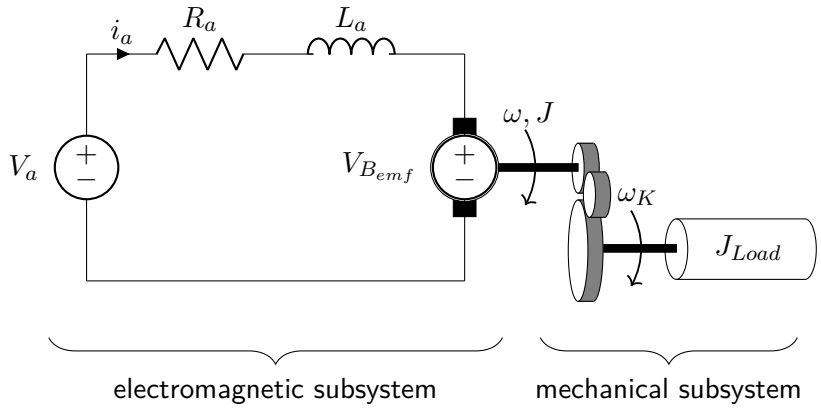
# 6 Motor transfer function




**Equations of the motor (without load)**

- Electrical: From Kirchhoff's Voltage Law (KVL) to the armature circuit

$$\begin{aligned} V_a(t) &= R_a i_a(t) + L_a \frac{\mathrm{d}i_a(t)}{\mathrm{d}t} + V_{B_{emf}}(t) \\ &= R_a i_a(t) + L_a \frac{\mathrm{d}i_a(t)}{\mathrm{d}t} + K_b \omega(t) \end{aligned} \tag{4}$$

- Mechanical: From Newton's second law to the rotor

$$\begin{aligned} J\frac{\mathrm{d}\omega(t)}{\mathrm{d}t} &= T_m(t) - b\omega(t) \\ &= K_t i_a(t) - b\omega(t) \end{aligned} \tag{5}$$

**Tabella 2:** DC Motor Variables and Parameters

| Symbol | Description | Units |
|---|---|---|
| $V_a(t)$ | Armature voltage (Input voltage) | V |
| $i_a(t)$ | Armature current | A |
| $R_a$ | Armature resistance | Ω |
| $L_a$ | Armature inductance | H |
| $V_b(t)$ | Back EMF (Electromotive Force) | V |
| $K_b$ | Back EMF constant | V/(rad/s) |
| $\omega(t)$ | Angular velocity of the rotor | rad/s |
| $T_m(t)$ | Motor torque developed by the motor | Nm |
| $K_t$ | Torque constant | Nm/A |
| $J$ | Moment of inertia of rotor and load | kg·m$^2$ |
| $b$ | Viscous friction coefficient (damping) | Nm/(rad/s) |

Let cosider the conservation of energy on the system in the steady state, where the electrical power is equal to the mechanical power:

- Electrical: electrical power in $= V_a i_a$

$$\begin{aligned} V_a &= i_a R_a + V_{B_{emf}} \\ &= i_a R_a + K_b \omega \end{aligned} \tag{6}$$

  with $K_b$ being the back EMF constant. So the electrical power in is:

$$\begin{aligned} P_{in} &= V_a i_a \\ &= i_a(i_a R_a + K_b \omega) \\ &= i_a^2 R_a + K_b i_a \omega \end{aligned} \tag{7}$$

If we consider an ideal motor without electrical losses, i.e. $R_a = 0$, then the electrical power in is:

$$P_{in} = K_b i_a \omega \tag{8}$$

- Mechanical: mechanical power out $= T_m \omega$ supposing no losses in the mechanical system.

$$\begin{aligned} P_{out} &= T_m \omega \\ &= K_t i_a \omega \end{aligned} \tag{9}$$

with $K_t$ being the torque constant.

For the conservation of energy, we have:

$$P_{in} = P_{out} \Rightarrow K_b i_a \omega = K_t i_a \omega \Rightarrow K_b = K_t = K \tag{10}$$

Using the Laplace transform on (4) and (5), we can derive the transfer function of the motor

$$G(s) = \frac{\Omega(s)}{V_a(s)} = \frac{K}{L_a J s^2 + (R_a J + L_a b)s + (R_a b + K^2)} \tag{11}$$

# 7 Estimation of motor parameters

To estimate the motor parameters, we can use the steady-state data collected from the motor under different PWM duty cycles i.e. at different voltages, currents, and speeds.

## 7.1 Torque and Back EMF Constants

As explained in the previous section, the torque constant $K_T$ and the back EMF constant $K_E$ can be considered equal for a DC motor. They can be estimated from the steady-state data stating from equation (4) at steady state which corresponds to (6):

$$V_a = i_a R_a + K_E \omega \rightarrow K_E = \frac{V_a - i_a R_a}{\omega} \tag{12}$$

Step by step:

1. Apply voltage $V_a$
2. let the motor run unloaded until it reaches a steady state
3. measure the armature current $i_a$ and the angular velocity $\omega$
4. calculate the back EMF constant $K_E$ using (12)
5. The torque constant $K_T$ is equal to $K_E$ in SI units.

## 7.2 Viscous Friction Coefficient

The viscous friction coefficient $b$ can be estimated from the steady-state data as follows:

$$b = \frac{K_T i_a}{\omega} \tag{13}$$

To estimate the viscous friction coefficient $b$, follow these steps:

1. Apply a voltage $V_a$ to the motor
2. Let the motor run unloaded until it reaches a steady state
3. Measure the armature current $i_a$ and the angular velocity $\omega$
4. Calculate the viscous friction coefficient $b$ using (13)

### 7.3 Moment of inertia $J$

The moment of inertia $J$ can be estimated in two ways:

**Method 1: From Coast-Down Experiment** Step-by-Step:

1. Run the motor at a known initial speed, $\omega_0$.
2. Cut the power and let the motor coast to a stop.
3. Measure the angular velocity over time, $\omega(t)$.
4. Use the governing differential equation's solution. The speed decays exponentially. By taking the natural logarithm, we linearize the equation:

$$\omega(t) = \omega_0 \cdot e^{-\frac{b}{J}t} \quad \Rightarrow \quad \ln(\omega(t)) = \ln(\omega_0) - \frac{b}{J}t$$

5. Plot $\ln(\omega)$ vs. $t$ and find the slope of the best-fit line. The slope directly relates to the system parameters:

$$\text{slope} = -\frac{b}{J} \quad \Rightarrow \quad J = -\frac{b}{\text{slope}}$$

**Note:** ✓You need to know (or have already measured) the viscous friction coefficient, $b$.

**Method 2: Step Response (System Identification)** Apply a known step torque (usually via a step current) and record the motor's initial angular acceleration. This method relies on Newton's second law for rotation, $T = J\dot{\omega}$.

$$T = J \cdot \dot{\omega} \quad \Rightarrow \quad J = \frac{T}{\dot{\omega}} = \frac{K_T \cdot I}{\dot{\omega}}$$

Step by step:

1. Apply a known step current, $I$, to the motor.
2. Measure the resulting initial angular acceleration, $\dot{\omega}$ (the slope of the speed graph right after the step is applied).
3. Calculate the inertia using the motor's torque constant, $K_T$:

$$J = \frac{K_T \cdot I}{\dot{\omega}}$$

## 8 Steady-State Data Collection and Analysis

Let's define the baseline parameters:

- $V_{\text{supply}} = 9.96\,\text{V}$ (Supply Voltage, measured)
- $R_a = 2.2\,\Omega$ (Armature Resistance, measured)
- PPR $= 3840$ (Encoder Pulses Per Revolution, known from STM32 timer configuration)
- $t_s = 0.01\,\text{s}$ (Sample Time, known from the STM32 timer configuration)
- $I_{\text{baseline}} = 67.6\,\text{mA}$ (Current sensor reading at no load/zero current, measured: comprise the current to drive the electronics and the motor at rest)

Table 3 summarizes the steady-state data collected from the DC motor at different PWM duty cycles. The data includes the applied voltage, measured current, speed, motor current, motor constants, and friction coefficient.

**Tabella 3:** Steady-State DC Motor Characterization Data

| Duty (%) | diffcount | Current (mA) | Voltage (V) | Speed (rad/s) | Motor I (A) | Motor Const. K ($\frac{Vs}{rad}$) | Friction b ($\frac{Nms}{rad}$) |
|---|---|---|---|---|---|---|---|
| 10 | 14 | 125.4 | 0.996 | 2.2896 | 0.0578 | 0.379 475 16 | 0.009 579 76 |
| 15 | 33 | 165.8 | 1.494 | 5.3969 | 0.0982 | 0.236 796 29 | 0.004 308 68 |
| 20 | 57 | 197.0 | 1.992 | 9.3219 | 0.1294 | 0.183 151 99 | 0.002 542 39 |
| 25 | 71 | 236.0 | 2.490 | 11.6115 | 0.1684 | 0.182 536 93 | 0.002 647 32 |
| 30 | 80 | 249.0 | 2.988 | 13.0833 | 0.1814 | 0.197 879 24 | 0.002 743 59 |
| 40 | 93 | 292.0 | 3.984 | 15.2094 | 0.2244 | 0.229 484 78 | 0.003 385 83 |
| 50 | 104 | 308.0 | 4.980 | 17.0083 | 0.2404 | 0.261 702 30 | 0.003 698 97 |
| 60 | 107 | 335.0 | 5.976 | 17.4990 | 0.2674 | 0.307 888 04 | 0.004 704 81 |
| | | | | **Mean (Media)** | | 0.247 364 34 | 0.004 201 42 |
| | | | | **Median (Mediana)** | | 0.233 140 53 | 0.003 542 40 |

The columns are calculated as follows:

- Applied Voltage ($V_a$): This is the average voltage applied to the motor via Pulse-Width Modulation (PWM). $V_a = \frac{\text{duty (\%)}}{100} \cdot V_{\text{supply}}$
- Speed ($\omega$): The angular velocity in radians per second is calculated from the encoder counts ('diffcount') measured over the sample time $t_s$. $\omega\,[\text{rad/s}] = \frac{\text{diffcount} \cdot 2\pi}{\text{PPR} \cdot t_s}$
- Motor Current ($I_a$): This is the current flowing through the motor, calculated by subtracting the baseline/offset from the raw sensor reading. $I_a\,[\text{A}] = \frac{\text{current}_{\text{raw}} - I_{\text{baseline}}}{1000}$
- Motor Constant ($K_E, K_T$): For a DC motor, the back-EMF constant ($K_E$ in V · s/rad) and the torque constant ($K_T$ in N · m/A) are numerically equal in SI units. At steady-state speed, the applied voltage equals the voltage drop across the resistor plus the back-EMF. $V_a = I_a R_a + E_{\text{back}}$ where $E_{\text{back}} = K_E \cdot \omega$ Solving for $K_E$ gives: $K_E = K_T = K = \frac{V_a - I_a R_a}{\omega}$
- Viscous Friction ($b$): At steady-state speed with no external load, the motor torque equals the friction torque. We model friction torque as being proportional to speed ($T_{\text{friction}} = b \cdot \omega$). $T_{\text{motor}} = K_T \cdot I_a = b \cdot \omega$ Solving for $b$ gives: $b = \frac{K_T \cdot I_a}{\omega}$

**Tabella 4:** Estimated Motor Parameters

| Symbol | Description | Units | |
|---|---|---|---|
| $R_a$ | Measured | 2.2 | $\Omega$ |
| $L_a$ | Measured | $2.3 \cdot 10^{-3}$ | H |
| $K$ | Estimated | 0.23 | V/(rad/s) |
| $J$ | Estimated by trial and error | 0.084 | kg·m$^2$ |
| $b$ | Estimated | 0.0035 | Nm/(rad/s) |

The transfer function of the system is given as a second-order model with two real poles:

$$
\begin{aligned}
G(s) &= \frac{0.25}{1.848 \cdot 10^{-5} s^2 + 0.01849 s + 0.07174} \\
&= \frac{1.353 \cdot 10^4}{s^2 + 1001 s + 3882} \\
&= \frac{1.353 \cdot 10^4}{(s + 996.6)(s + 3.89)} \\
&= \frac{3.4848}{(1 + s\frac{1}{996.60})(1 + s\frac{1}{3.89})} \\
&\approx \frac{3.4848}{(1 + s0.001)(1 + s0.2567)}
\end{aligned}
\tag{14}
$$

As shown, the system has two real poles located at $s = -996.6$ and $s = -3.89$. These correspond to two time constants: a fast dynamic with time constant approximately $1/996.6 \approx 1ms$, and a slow dynamic with time constant $1/3.89 \approx 0.2567$s. Since the pole at $s = -3.89$ lies much closer to the origin (to the imaginary axis), it is the dominant pole, and it governs the system's overall dynamic behavior at lower frequencies.

We can therefore simplify the system using a first-order approximation, by neglecting the faster pole. The reduced transfer function becomes:

$$
G_{\text{approx}}(s) \approx \frac{3.4848}{1 + 0.2567s} \tag{15}
$$

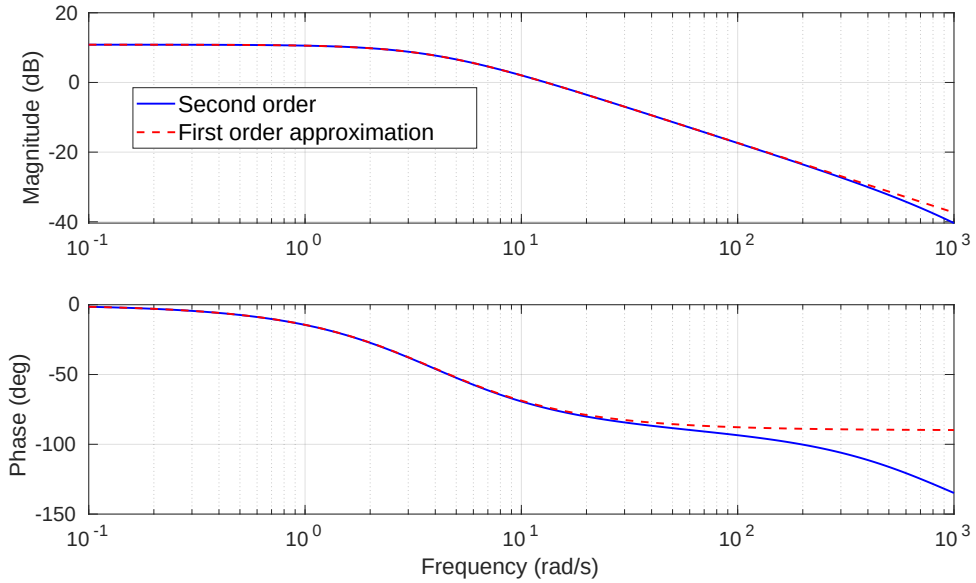



**Figura 31:** Bode plot of the estimated transfer function compared to the first order approximation.

This approximation retains the dominant dynamics while significantly simplifying the model. The validity of this approximation can be assessed by comparing the frequency responses, as shown in Figure 31. The Bode plot illustrates that, up to a certain frequency (well below the corner frequency of the neglected pole at approximately $10^3 \text{rad}\,\text{s}^{-1}$), the magnitude and phase responses of the original and reduced models are nearly indistinguishable. This confirms that the first-order model accurately captures the essential system behavior in the frequency range of interest.

# 9 First Order Transfer Function Approximation

Let us consider the first-order transfer function:

$$G(s) = \frac{\mu}{1+Ts} \quad \left[\frac{\text{rad/s}}{V}\right]$$

where:

- $\mu$ is the static gain of the system,
- $T$ is the time constant.

Our goal is to estimate the parameters $\mu$ and $T$ based on the system's response to a step input.

Suppose we apply a generic step input of amplitude $V_a$. The Laplace transform of the input is $V_a/s$, and the output in the Laplace domain becomes:

$$F(s) = G(s) \cdot \frac{V_a}{s} = \frac{V_a}{s} \cdot \frac{\mu}{1+Ts} \tag{16}$$

### Estimation of the Gain $\mu$

To find the steady-state value of the system response, we apply the Final Value Theorem (FVT):

$$f_\infty = \lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} V_a \cdot \frac{\mu}{1+Ts} = V_a \cdot \mu \tag{17}$$

Where:

- $V_a$ is the known amplitude of the step input,
- $f_\infty$ is the steady-state value of the system output (measurable from the response),
- $\mu$ is the gain we want to estimate.

Solving for $\mu$ gives:

$$\mu = \frac{f_\infty}{V_a} \tag{18}$$

### Estimation of the Time Constant $T$

To estimate the time constant $T$, we perform the inverse Laplace transform of (16):

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \mu V_a \left(1 - e^{-\frac{t}{T}}\right) \tag{19}$$

Evaluating this expression at $t = T$:

$$f(T) = \mu V_a \left(1 - e^{-1}\right) \approx 0.632 \cdot \mu V_a$$

Since $\mu = \frac{f_\infty}{V_a}$, we substitute:

$$f(T) = 0.632 \cdot f_\infty$$

This result indicates that the time constant $T$ corresponds to the time required for the system response to reach 63.2% of its final value. Hence, knowing $f_\infty$, we can determine $T$ as the time at which $f(t) = 0.632 \cdot f_\infty$.

**Example 11** Consider a DC motor to which a step input voltage of $V_a = 5\,\mathrm{V}$ is applied. The corresponding step response is shown in Figure 32.
From the plot, the steady-state output is observed to be:

$$f_\infty = 4.58\,\frac{\mathrm{rad}}{\mathrm{s}}$$

Given the applied voltage $V_a = 5\,\mathrm{V}$, the system gain is estimated as:

$$\mu = \frac{f_\infty}{V_a} = \frac{4.58\,\mathrm{rad/s}}{5\,\mathrm{V}} = 0.916\,\frac{\mathrm{rad/s}}{\mathrm{V}}$$

To determine the time constant, compute 63.2% of the final value:

$$f(T) = 0.632 \cdot f_\infty = 0.632 \cdot 4.58\,\mathrm{rad/s} = 2.89\,\mathrm{rad/s}$$

By examining the step response, we find that this value is reached at $t = 1.58\,\mathrm{s}$. Therefore:

$$T = 1.58\,\mathrm{s}$$

The resulting estimated transfer function of the system is:

$$G(s) = \frac{0.916}{1 + 1.58s} \quad \left[\frac{\mathrm{rad/s}}{\mathrm{V}}\right]$$

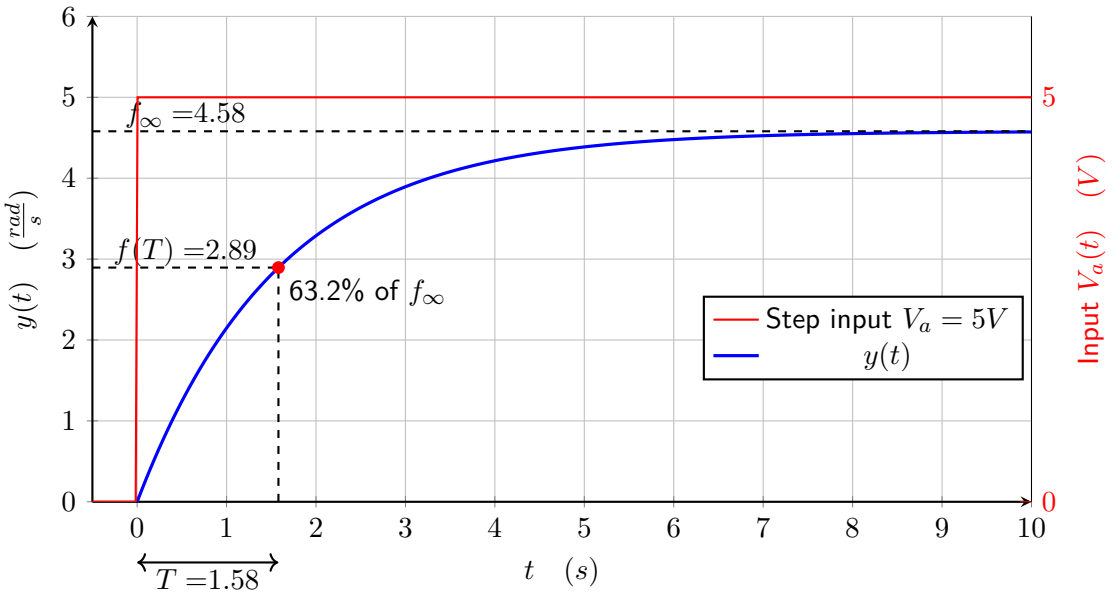



**Figura 32:** Step response of the system with estimated parameters. The red point marks the inflection point where the response reaches 63.2% of the final value.

## 10 First-Order Plus Time Delay (FOPTD) Transfer Function

Consider the step response shown in Figure 33, where the output exhibits a time delay of $\tau$ seconds before any noticeable change occurs following the step input. This behavior is characteristic of systems that can be modeled by a first-order plus time delay (FOPTD) transfer function, expressed as:

$$P(s) = \frac{\mu}{1 + Ts} \cdot e^{-\tau s} \quad \left[\frac{\mathrm{rad/s}}{V}\right]$$

In this expression:

- $\mu$ represents the steady-state gain of the system,
- $T$ is the time constant,
- $\tau$ is the pure time delay (or dead time).

To estimate the parameters $\mu$, $T$, and $\tau$, one can adopt a graphical method based on the step response as seen in the previous section. The time delay $\tau$ is determined as the interval between the step input and the point where the output begins to rise. The time constant $T$ is then measured from the onset of the response (i.e., from $t = \tau$) to the time at which the output reaches 63.2% of its final steady-state value. This method assumes a monotonic response and is particularly useful for systems where the delay significantly influences the dynamics.

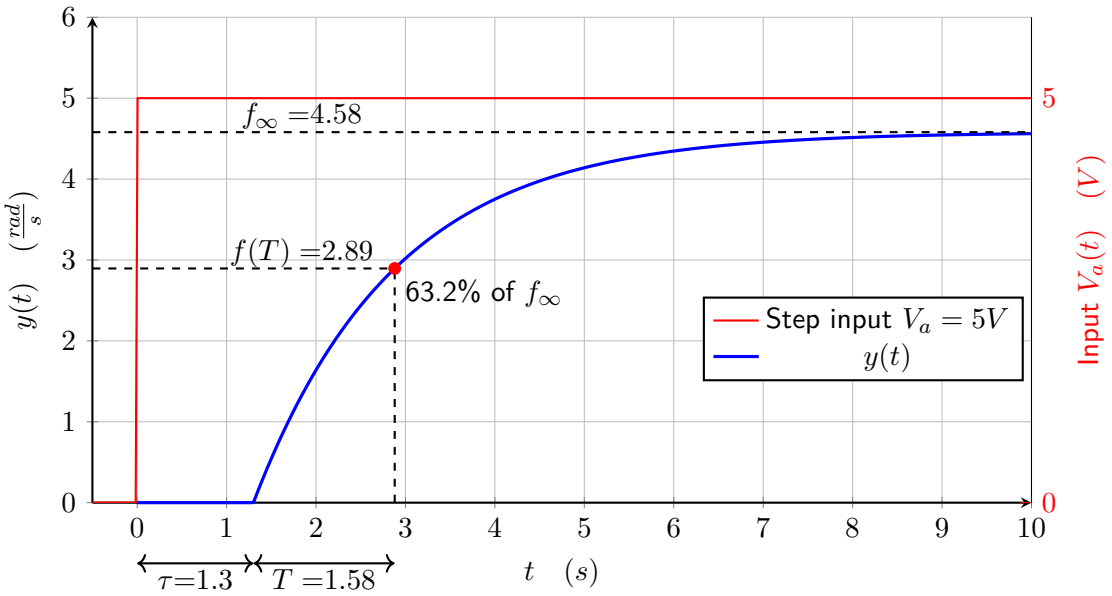



**Figura 33:** Step response of the system. The red line and right-hand axis show the input voltage $V_a(t)$, while the blue line and left-hand axis show the angular velocity response $y(t)$. The system parameters are estimated from the response, with the red point marking the inflection point where the response reaches 63.2% of the final value.

### 10.0.1 Tangent method

The tangent method is an alternative technique used to estimate the parameters of a FOPDT system directly from its step response.

**Procedure:** With reference to Figure 34

1. Apply a known input step to the system and record the output $y(t)$.
2. Identify the final steady-state value of the output, denoted as $f_\infty$.
3. Compute the gain $\mu$ according to (18)
4. Calculate the time derivative of the response $\dot{y}(t)$. Since the response is sampled, this can be approximated by calculating the numerical derivative

$$\dot{y}(k) \approx \frac{y(k+1) - y(k)}{t(k+1) - t(k)}$$

   where $k$ is the sample index.

5. Determine the point $(t_m, y(t_m))$ of **maximum slope** of the curve, i.e. the time $t_m$ where $\dot{y}(t)$ reaches its maximum value $\dot{y}_{\max}$.

$$t_m = \arg\max_t \dot{y}(t), \qquad y(t_m) \text{ is the value of the function at } t_m.$$

6. Draw the tangent line to the curve at $(t_m, y(t_m))$. The equation of the tangent is:

$$y(t) = \dot{y}_{\max}(t - t_m) + y(t_m)$$

7. Find the time $t_0$ where the tangent intersects the time axis

$$t_0 = t_m - \frac{y(t_m)}{\dot{y}_{\max}}$$

8. The tangent intersects the final steady-state value $y_\infty$ at time:

$$t_1 = t_m + \frac{y_\infty - y(t_m)}{\dot{y}_{\max}}$$

9. The system parameters are then estimated as:

$$\tau = t_0, \qquad T = t_1 - t_0$$

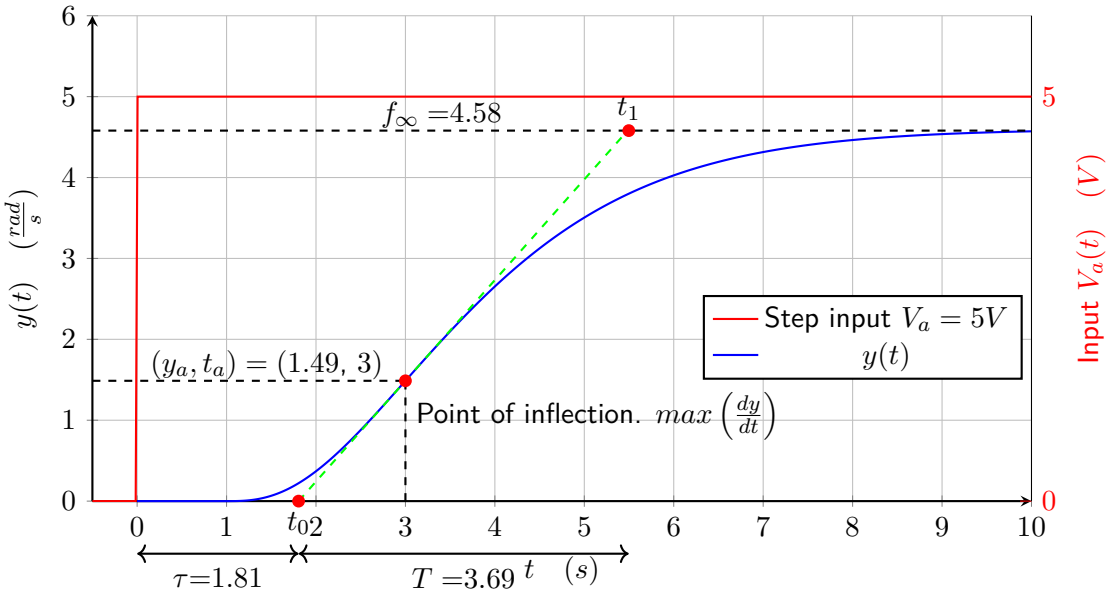



**Figura 34:** Example of the Tangent Method for estimating the parameters of a first-order system with delay from its step response. The red line represents the step input $V_a(t)$, while the blue line shows the system output $y(t)$. The green dashed line is the tangent at the point of inflection (maximum slope) of the output curve. The intersections of this tangent with the time axis and the steady-state output level allow for the estimation of the delay $\tau$ and time constant $T$ of the system.

**Effect of the noise:** In the presence of measurement noise, the identification process may yield less accurate estimates of the parameters $\mu$, $T$, and $\tau$. Especially for the calculation of the tangent at the inflection point, which requires the derivative of the output signal, noise can significantly affect the slope estimation. Consider for example Figure 35, which illustrates how noise can distort the tangent line and lead to incorrect parameter estimates.

In this example, the estimated parameters for the three cases are:

- Original: $\tau = 1.8054 \quad T = 3.6920$
- Noisy: $\tau = 5.4802 \quad T = 0.2250$
- Filtered (LPF 1Hz): $\tau = 2.3248 \quad T = 2.9173$

To mitigate this, it is advisable to apply filtering techniques to the measured output data before performing the identification. However, as can be seen in Figure 35, filtering can introduce additional delays and alter the system dynamics, it is important to choose the filter parameters carefully to balance noise reduction and signal integrity.

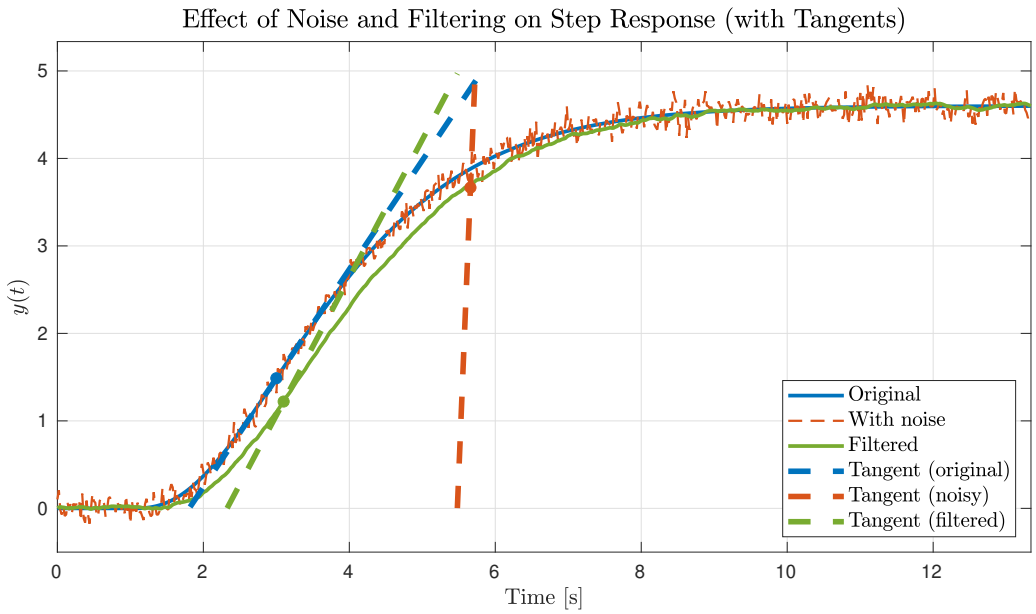



**Figura 35:** Effect of measurement noise on the Tangent Method for system identification

# 11 Selection of the Input Step Amplitude

Assuming an idealized linear behavior—neglecting non-linearities and non-ideal effects—it is expected that the choice of input step amplitude will influence the resulting system response and, consequently, the estimated parameters $\mu$, $T$, and $\tau$ (this can be clearly seen in Table 3). For this reason, it is advisable to identify the system parameters at or near the actual operating point of the final application.

In the following, we consider two simple strategies for selecting the input step amplitude.

### Hypothesis 1: No Prior Knowledge of the Operating Point

Suppose that no prior information is available about the final application—neither the expected operating speed nor the expected input voltage. In such cases, a straightforward approach is to estimate the system parameters at several operating points, sweeping the step amplitude from zero up to the maximum allowable motor voltage.

For each step amplitude, the system's response is recorded and the parameters $\mu$, $T$, and $\tau$ are estimated. An average of the resulting parameter values can then be computed to obtain a representative model. To improve robustness, a weighted average may be used, assigning less weight to estimates at the extremes of the operating range—i.e., near zero voltage (where effects such as stiction ) and near the maximum voltage (where saturation and supply limitations may dominate).

### Hypothesis 2: Known Operating Point

If the expected operating conditions are known—such as the nominal speed or voltage during the final application—then the input step amplitude can be selected accordingly. In this case, the input voltage should be chosen to reflect typical real-world conditions, and the parameters $\mu$, $T$, and $\tau$ should be estimated at that point.

### Example: Line-Following Robot

As an illustrative example, consider the final goal of the course: to control a robot to follow a line. Based on the layout of the course map, we observe that approximately 60% of the path consists of straight lines, while the remaining 40% includes curves and corners.

On straight segments, the robot is expected to travel at its maximum speed, corresponding to the maximum voltage supported by the motor, denoted as $V_{\text{motor}_{\text{max}}}$. In corners, the speed must be reduced—let us conservatively assume that the robot travels at only 10% of its maximum speed (and thus receives 10% of the maximum motor voltage).

A reasonable choice for the input step amplitude $V_a$ is to use a weighted average based on the relative time spent in each condition:

$$V_a = 0.6 \cdot V_{\text{motor}_{\text{max}}} + 0.4 \cdot 0.1 \cdot V_{\text{motor}_{\text{max}}} = 0.64 \cdot V_{\text{motor}_{\text{max}}}$$

This value can then be used as the input step amplitude for the identification process. Estimating the system parameters at this representative operating point provides a more realistic model for the expected behavior of the robot during its task.

# 12 Tuning of a PI Controller for the DC Motor with Ziegler-Nichols Method

The Ziegler-Nichols method is a heuristic approach for tuning PID controllers based on the system's response to a step input. It allows to determine the parameters for a PID controller in the form

$$\begin{aligned} C(s) &= K_p \left(1 + \frac{1}{T_i s} + T_d s\right) \\ &= K_p + \frac{K_i}{s} + K_d s \quad \text{with } K_i = \frac{K_p}{T_i}, \quad K_d = K_p T_d \end{aligned} \tag{20}$$

## 12.1 Method 1: Open-Loop

Using the step response of the open-loop system, we can estimate the parameters $\mu$, $T$, and $\tau$ using the Tangent Method as described earlier. Once these parameters are known, the Ziegler-Nichols tuning rules for a PID controller can be applied as follows:

| **Type of Controller** | $K_p$ | $T_i$ | $T_d$ |
|---|---|---|---|
| P | $\frac{1}{\mu}\frac{T}{\tau}$ | – | – |
| PI | $\frac{0.9}{\mu}\frac{T}{\tau}$ | $3\tau$ | – |
| PID | $\frac{1.2}{\mu}\frac{T}{\tau}$ | $2\tau$ | $0.5\tau$ |

**Tabella 5:** Ziegler–Nichols open-loop tuning parameters

**Note:** This method works only when the system has an input delay $\tau > 0$.

## 12.2 Method 2: Closed-Loop

Alternatively, the Ziegler-Nichols method can be applied using the closed-loop response of the system. This involves setting up a pure proportional feedback loop and gradually increasing the proportional gain $K_p$ until the stability boundary where the system output exhibits sustained oscillations. The gain at which this occurs is called the critical gain $K_{cr}$, and the period of the oscillations is called the critical period $T_{cr}$ (Figure 36).

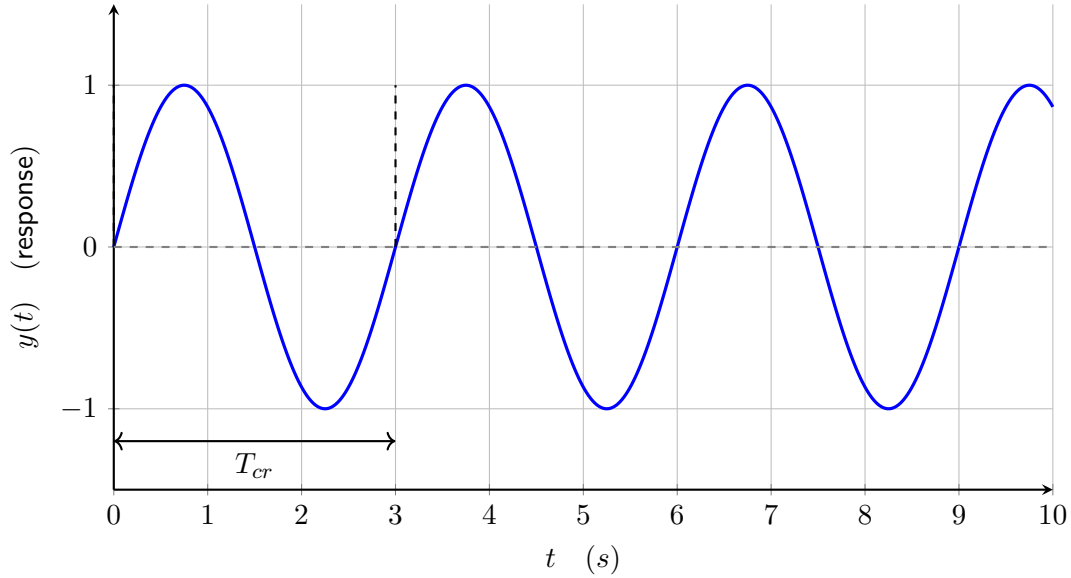


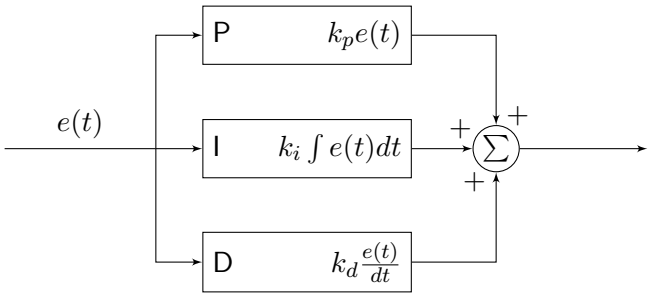

**Figura 36:** Sustained oscillation at the critical gain $K_{cr}$ used in the Ziegler–Nichols closed-loop tuning method. The period of oscillation is the critical period $T_{cr}$.

The Ziegler-Nichols tuning rules for a PID controller based on the critical gain and period are as in Table 6.

**Tabella 6:** Ziegler–Nichols Closed-Loop (Critical Gain) Tuning Rules

| **Controller Type** | $K_p$ | $T_i$ | $T_d$ |
|---|---|---|---|
| P | $0.5\,K_{cr}$ | – | – |
| PI | $0.45\,K_{cr}$ | $T_{cr}/1.2$ | – |
| PD | $0.8\,K_{cr}$ | – | $0.125\,T_{cr}$ |
| PID | $0.6\,K_{cr}$ | $0.5\,T_{cr}$ | $0.125\,T_{cr}$ |

# 13 PID implementation




We need to implement the discrete version of the PID controller.

```
1
2  float PI(float error)
3  {
4      float P =  kp * error;
5      static float I = 0;
6      I = I + ki * error * TS;
7
8      return P + I;
9  }
```